Ex.No.6

31-07-2026

FIND MEAN,MEDIAN, MODE FOR THE GIVEN SET OF NUMBERS IN A LIST

AIM

To find mean, median, mode for the given set of numbers in a list.

ALGORITHM

Step 1: Start the program.
Step 2: Initialize a list of numbers
num = [10, 20, 20, 30, 40]
Step 3: Calculate the **Mean** by dividing the sum of all elements by the total number of elements.
Step 4: Sort the list in ascending order.
Step 5: Find the **Median**

- If the number of elements is even, calculate the average of the two middle elements.
- If the number of elements is odd, select the middle element.

Step 6: Find the **Mode** by selecting the element that occurs most frequently in the list.
Step 7: Display the list, Mean, Median, and Mode, then stop the program.

PROGRAM

```
num = [10, 20, 20, 30, 40]

# Mean

mean = sum(num) / len(num)

# Median

num.sort()

n = len(num)

if n % 2 == 0:

    median = (num[n//2 - 1] + num[n//2]) / 2

else:

    median = num[n//2]

# Mode

mode = max(num, key=num.count)

print("Numbers:", num)

print("Mean:", mean)

print("Median:", median)
```

```
print("Mode:", mode)
```

OUTPUT

RESULT:

Thus, the program to find mean, median, mode for the given set of numbers in a lists is executed successfully.